Venkata Lakshmi Parimala Pasupuleti

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PROFESSIONAL SUMMARY

AI Forward Deploy Engineer with 8+ years turning technical solutions into adopted, measurable business outcomes. I run the full lifecycle -- discovery sessions with business leaders and operations teams, solution architecture, rapid prototyping, custom API integration, production deployment, user training, and adoption measurement. At Entrata, I was the embedded AI and systems engineer for enterprise operational workflows at 100K+ scale: I conducted discovery with legal and operations stakeholders, translated complex business requirements into technical designs, built integrations across 5+ dependent platforms, deployed scalable capabilities into production, and owned the outcomes. Now building production agentic AI: LeaseI is an 8-agent LangGraph pipeline with AI guardrails, responsible AI controls, and HITL escalation (93.6% precision); AgentProbe ships the eval harness that validates it. Python, FastAPI, LangChain, LangGraph, cloud platforms (AWS/Azure/GCP), and enterprise data integration -- production-grade. M.S. Data Analytics Engineering, GMU (GPA 3.87). AgentProbe: github.com/venkatalakshmiparimala/AgentProbe

CORE COMPETENCIES

AI Solution Deployment & Adoption | Discovery & Requirements Translation | Rapid Prototyping & Proofs of Concept | Enterprise API Integration | Business Process Transformation | Executive Stakeholder Communication | Agentic AI & LangGraph | AI Guardrails & Responsible AI

WORK EXPERIENCE

Blue Horse Digital

Sep 2026 -- Present

AI/ML Engineer (Volunteer) -- Virginia, USA

- Building ML models and agentic AI pipelines for business applications; integrating data systems with back-end APIs; measuring solution effectiveness and iterating based on operational behavior and team feedback.

Magizhs Tech Solution

May 2025 -- Aug 2025

AI/ML Engineer Intern (Forward Deployed)

- **Ran the full forward deploy lifecycle -- discovery through adoption:** conducted discovery sessions with marketing and real estate operations teams to understand workflows and desired outcomes; translated business requirements into a LangGraph agentic solution (parse → plan → retrieve → validate → routing); built custom integrations with enterprise data sources; deployed into production; trained users on the deployed AI capability.
- Integrated RAG (FAA Part 107 corpus) for grounded, auditable compliance decisions; engineered AI guardrails blocking non-compliant outputs before they reached operational users; measured and reported solution effectiveness to business stakeholders.

George Mason University

Aug 2024 -- May 2026

Graduate Teaching Assistant + Research Associate

- Designed and delivered technical training to 600+ students across multiple sections; provided one-on-one troubleshooting support; adapted instructional approaches based on adoption signals and comprehension outcomes.
- Built multi-source prediction pipelines (Python, R, $R^2 = 0.977$); designed Tableau/Quarto dashboards for faculty leadership translating complex analytical findings into executive-ready insights (+40% comprehension).

Entrata (Enterprise PropTech SaaS)

Jan 2017 -- Aug 2024

Senior Software Engineer -- Lehi, UT (Remote) [promoted from Associate, 2018]

- **Acted as the embedded technical partner for enterprise operational deployments** -- conducted discovery with business leaders, legal, and operations teams to understand workflows and regulatory requirements; designed solution architectures integrating AI and data systems across 5+ dependent platforms; deployed scalable capabilities into production for 100K+ monthly active users; drove business adoption across operational teams. Zero compliance violations and zero operational incidents across 7 years.
- **Built and deployed custom integrations across enterprise systems:** designed RESTful APIs and microservices connecting internal platforms, third-party data sources, and cloud services (AWS S3 migration with 100% PHI compliance); built async pipelines (RabbitMQ) for large-scale data processing with idempotency and retry guarantees; eliminated 80% of manual operational workload through automation -- measurable operational impact.
- Translated complex regulatory requirements (HUD/LIHTC) into technical designs and communicated implementation progress to executive and legal stakeholders; supported process transformation by helping operations teams incorporate new systems into daily workflows; optimized 200+ SQL queries (30% performance gain); mentored 5+ engineers via 300+ code reviews.

PROJECTS

LeaseI: Agentic AI Investigation Platform with Responsible AI Controls

LangGraph 8-agent -- HITL -- FastAPI -- RAG -- AI Guardrails

Production agentic AI solution deployed into an operational workflow -- moving from prototype to production with measurable business outcomes. Architecture: 8-agent LangGraph pipeline (discovery → retrieval → configuration analysis → historical incident review → structured data → reasoning → validation → human escalation); RAG over operational documents with citation grounding; responsible AI controls enforcing that AI never auto-resolves regulated decisions (always routes to human); AI guardrails blocking unsupported outputs before they reach users. Results: 93.6% precision / 87.9% F1 across 1,240 test cases; 100% citation correctness on Phase 1 eval; zero unsupported AI decisions reaching operational users.

github.com/venkatalakshmiparimala/Lease_complaince_rag

AgentProbe: AI Solution Eval & Observability Harness

LangGraph -- LangChain -- FastAPI -- ChromaDB -- CI/CD

Production eval harness for measuring AI solution effectiveness -- the instrumentation layer that validates whether deployed AI solutions are achieving their operational objectives. 15-endpoint FastAPI backend, 5-metric LLM-as-judge engine (correctness, groundedness, latency, token efficiency, hallucination rate), CI/CD regression gate, multi-provider LLM router (OpenAI/Anthropic/Groq/Ollama), observability dashboard tracking solution performance over time. Provides the measurement framework for evaluating progress against defined business objectives.

github.com/venkatalakshmiparimala/AgentProbe

AI on FHIR: NLP-Driven Clinical Records Query System

Flask -- LLMs -- BERT -- React -- Docker -- FHIR

Scoped with clinical operations staff, designed, built, and deployed an NLP platform translating natural-language queries (EN/ES) into FHIR API calls using LLMs, BERT, and embeddings; HIPAA-compliant RBAC; trained staff on the deployed AI capability; drove adoption across clinical workflows. Measurable business outcome: ~60% reduction in manual retrieval time.

github.com/venkatalakshmiparimala/AI_on_FHIR

EDUCATION

M.S. Data Analytics Engineering -- George Mason University — May 2026

GPA 3.87 -- Virginia, USA

B.Tech Computer Science & Engineering -- JNTU Anantapur — 2017

SKILLS

AI & Agents: LangChain, LangGraph, Agentic AI, LLMs, Generative AI, RAG Pipelines, Prompt Engineering, Function Calling, AI Guardrails, Responsible AI, Content Filtering, HITL, Safety Controls, Multi-Agent Orchestration, Observability, Hallucination Detection, OpenAI API, Anthropic API, FastAPI, MCP

Integration & Deployment: REST APIs, Enterprise API Integration, Data Integration, Third-Party Integrations, Docker, Kubernetes, CI/CD (GitHub Actions), Microservices, AWS (S3, EC2, Lambda, Glue, Bedrock), Azure, Google Cloud Platform (GCP), RabbitMQ, PostgreSQL, Redis

Languages & Frameworks: Python, JavaScript, TypeScript, SQL, NoSQL, Flask, Django, React.js, Spring Boot, Java, Node.js, PHP

ML & Data: PyTorch, HuggingFace Transformers, Scikit-learn, XGBoost, Isolation Forest, Pandas, NumPy, Tableau, Solution Architecture, Business Process Transformation, Stakeholder Management, Executive Communication